

Multiple Choice (10 marks)

1. What is the significance of the 1:2:4 resonance in the Jupiter's moons system?
 - (a) The resonance pulls Io in different directions and generates heat.
 - (b) It makes the orbit of Io slightly elliptical.
 - (c) It creates a gap with no asteroids between the orbits.
 - (d) It prevents formation of the ring material into other moons.
2. Why do we see essentially the same face of the Moon at all times?
 - (a) because the Moon has a nearly circular orbit around the Earth
 - (b) because the Moon does not rotate
 - (c) because the other face points toward us only at new moon when we can't see the Moon
 - (d) because the Moon's rotational and orbital periods are equal
3. What drives differentiation?
 - (a) Spontaneous emission from radioactive atoms.
 - (b) The minimization of gravitational potential energy.
 - (c) Thermally induced collisions.
 - (d) Plate tectonics.
4. What are the conditions necessary for a terrestrial planet to have a strong magnetic field?
 - (a) fast rotation only
 - (b) a rocky mantle only
 - (c) a molten metallic core only
 - (d) both a molten metallic core and reasonably fast rotation
5. Which of the following lists the ingredients of the solar nebula from highest to lowest percentage of mass in the nebula?
 - (a) hydrogen compounds (H₂ O CH₄ NH₃) light gases (H He) metals rocks
 - (b) hydrogen compounds (H₂ O CH₄ NH₃) light gases (H He) rocks metals
 - (c) light gases (H He) hydrogen compounds (H₂ O CH₄ NH₃) metals rocks
 - (d) light gases (H He) hydrogen compounds (H₂ O CH₄ NH₃) rocks metals

True / False (10 marks)

Answer True or False

6. True or False: The black hole at the center of the Milky Way galaxy is named Alsephina A*.
7. True or False: A large baked potato takes longer to cool than a small baked potato due to its greater volume and surface area ratio.
8. True or False: The Moon's surface is less heavily cratered than the Earth's surface due to a lack of geological activity.
9. True or False: The Moon can be in the waning crescent phase when it sets at noon, as it is still visible during daylight hours.
10. True or False: Jupiter and the other jovian planets contain a significant fraction of non-gaseous matter, challenging the term 'gas giants.'

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the formation, characteristics, and significance of the Great Red Spot on Jupiter, and analyze its implications for our understanding of atmospheric dynamics in gas giants.
12. Discuss the implications of measuring astronomical distances, specifically calculating the distance to Altair, given it is one million times greater than the Earth-Sun distance.

End of Paper